PHYSICS **TARGET : JEE-ADVANCED-2024**	**CAPS-5+**

1. 1 mole of an ideal gas undergoes a process such that $P \propto \frac{1}{T}$. The molar heat capacity of this process is 33.24 J/mol K. Which of the following is **incorrect**

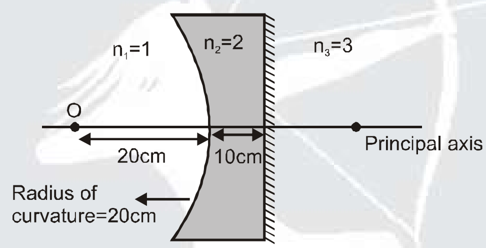
(A) The work done by the gas is $2R\Delta T$.

(B) $\gamma = \left(\frac{C_P}{C_V}\right)$ for the gas is 1.5.

(C) Degree of freedom of the gas is 3.

(D) None of the above

2. Final image of point object 'O' formed by the combination is located :



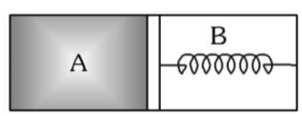
(A) On plane surface

(B) at a distance 10cm from plane surface

(C) at a distance 30cm from plane surface

(D) at a distance 20cm from curved surface

3. A thermally insulated chamber of volume $2V_0$ is divided by a frictionless piston of area S into two equal parts A and B. Part A has an ideal gas at pressure P_0 and temperature T_0 and in part B is vacuum. A massless spring of force constant K is connected with the piston and the wall of the container as shown. Initially the spring is unscratched. The ideal gas in chamber A is allowed to expand slowly. Let in equilibrium the spring be compressed by a length x_0, then



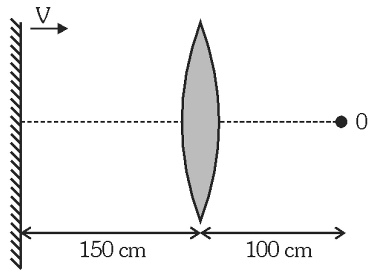
(A) Final pressure of the gas is Kx_0/S

(B) Work done by the gas is $\frac{1}{2}kx_0^2$

(C)Magnitude of change in internal energy of the gas is $\frac{1}{2}kx_0^2$

(D) None of these

4. An observer is standing at a point O, at a distance of 100 cm from a convex lens of focal length 50 cm. A plane mirror is placed behind the lens at a distance of 150 cm from the lens. The mirror now starts moving towards the right with a velocity of 10 cm/s. What will be the magnitude of velocity (in m/s) of her own image as seen by the observer, at the moment when the mirror just starts moving ?

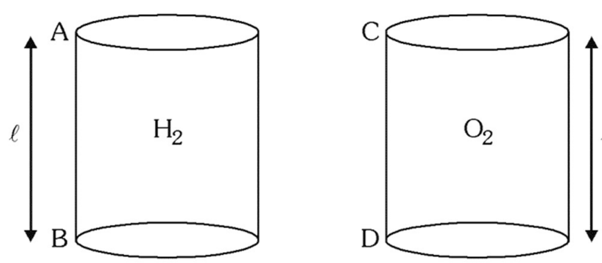


(A) $\frac{1}{9}\text{cm/s}$ (B) $\frac{10}{9}\text{cm/s}$ (C) $\frac{20}{9}\text{cm/s}$ (D) $\frac{20}{3}\text{cm/s}$

5. When the nucleus of $_7N^{14}$ is bombard with an α-particle of sufficient kinetic energy, it gets transformed into $_8O^{17}$. If the masses of $_7N^{14}$, $_2He^4$, $_1H^1$ and $_8O^{17}$ are respectively 14.00307u, 4.00260u, 1.00783u and 16.99913u and 1u = 931.5 MeV, if the minimum kinetic energy that must be possessed by the α particle is xyz × 10^{-2} MeV then find value of $\frac{x+y+z}{2}$.

(A) 5 (B) 10 (C) 15 (D) 20

6. Two cylindrical pipes are each of length $\ell = 30$ cm. One of them contains hydrogen and other has oxygen at same temperature. The ends A,B, C,D of pipes are fitted with flexible diaphragms. The diaphragm A and C are set into oscillation simultaneously using same source having frequency f = 600 Hz. The difference in phase of oscillation of the diaphragms D and B if it is known that the speed of sound in hydrogen at the temperature concerned is 1200 m/s.



(A) 0.3 π radian (B) 0.6 π radian (C) 1.2 π radian (D) 0.9 π radian

7. The kinetic energy of a particle continuously increases with time

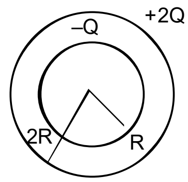
(A) the resultant force on the particle must be parallel to the velocity at all instants.

(B) the resultant force on the particle must be at an angle less than 90° with the velocity all the time

(C) its height above the ground level must continuously decrease

(D) the magnitude of its linear momentum is increasing continuously

8. Charge – Q and 2Q are distributed uniformly on surface of two concentric spherical shells of radii 'R' and '2R' respectively as shown in the figure. Select correct alternative(s)



(A) the total electrostatic energy stored in the system is $\frac{Q^2}{8\pi\varepsilon_0 R}$

(B) electrostatic energy in the space between two shells is $\frac{Q^2}{16\pi\varepsilon_0 R}$

(C) electrostatic energy stored outside the system is $\frac{Q^2}{2\pi\varepsilon_0 R}$

(D) electrostatic energy in space between two shells is zero.

9. A radioactive nucleus A decy to C after emitting two α and three β particles. Another nucleus B decays to C by emitting one α and five β particles. Half life of A and B are 1 hr and 2hr respectively and all intermediates have negligible half lives. At time t = 0, population of nuclei of A and B are $4N_0$ and N_0 respectively and population of C is zero.

(A) The difference in atomic numbers and mass numbers of A and B is 4

(B) The time t_0 when both A and B have equal rate of decay is 6 hr.

(C) The population of C when both A and B have equal population is $\frac{9N_0}{2}$

(D) The time t_0 when both A and B have equal rate of decay is 3 hr.

10. P.E. of a system of n positive point charges is 100 μJ. Out of these n charges two charges q_1 and q_2 have interaction energy = 1 μJ. The charge q_1 is removed without affecting other charges, now the potential energy of rest of system is 90 μJ. Now charge q_2 is also removed. The Potential energy of remaining system is 82 μJ. Choose the correct options

(A) Interaction potential energy of charge q_1 with rest of n – 2 charges (except q_2) is 9μJ.

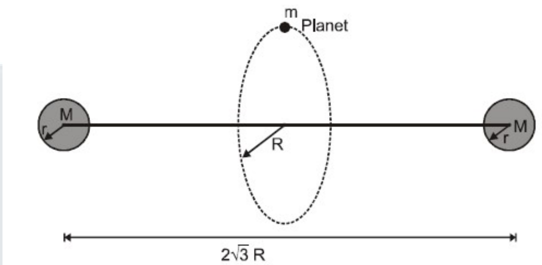
(B) Interaction potential energy of charge q_2 with rest of n – 1 charges is 9μJ.

(C) Interaction potential energy of charge q_1 with rest of n – 2 charges (except q_2) is 10μJ.

(D) Interaction potential energy of charge q_2 with rest of n – 1 charges is 8μJ.

COMPREHENSION

Consider a hypothetical solar system, which has two identical massive suns each of mass M and radius r, seperated by a seperation of $2\sqrt{3}R$ (centre to centre). (R >>>r). These suns are always at rest. There is only one planet in this solar system having mass m. This planet is revolving in circular orbit of radius R such that centre of the orbit lies at the mid point of the line joining the centres of the sun and plane of the orbit is perpendicular to the line joining the centres of the suns. Whole situation is shown in the figure.



Answer the following questions regarding to this solar system.

11. Average force on the planet in half revolution is :

(A) $\frac{GMm}{4\pi R^2}$ (B) $\frac{GMm}{4R^2}$ (C) $\frac{GMm}{2\pi R^2}$ (D) $\frac{GMm}{8R^2}$

12. Duration of one year for this planet is :

(A) $\frac{4\pi R^{3/2}}{\sqrt{GM}}$ (B) $\frac{2\pi R^{3/2}}{\sqrt{GM}}$ (C) $\frac{\pi R^{3/2}}{\sqrt{GM}}$ (D) $\frac{3\pi R^{3/2}}{\sqrt{GM}}$

13. A satellite revolves around a planet, in a certain orbit of radius r_0. If V_0, K_0, T_0, P_0, E_0, B_0 and L_0 represent physical quantities as given below then match the entries of Column I & Column II for increasing radius of orbit.

V_0 – orbital velocity

T_0 – Time period

K_0 – Kinetic Energy of Satellite.

P_0 – Potential Energy of Satellite.

E_0 – Total Energy of Satellite.

B_0 – Binding Energy of Satellite

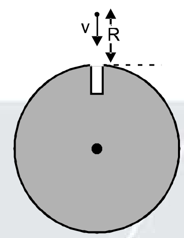
L_0 – Angular Momentum of Satellite about the centre of planet.

List-I	**List-II**
(P) V_0K_0	(1) Increase
(Q) B_0L_0	(2) Decrease
(R) $P_0E_0L_0^4$	(3) Independent of r
(S) $\frac{T_0}{L_0^3}$	(4) Proportional to $r^{-3/2}$

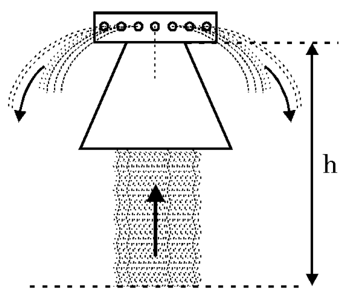
Code :

	P	**Q**	**R**	**S**
(A)	1, 2	1, 3	2	2
(B)	2, 4	2	3	3
(C)	2, 4	1, 4	3	2
(D)	1, 3	3	4	1

14. A fixed sphere of radius R and charge Q(uniformly) has a small groove of length $\frac{R}{4}$ as shown in figure. Another point charge of mass 'm' and charge 'q' is projected towards the groove with some velocity in given direction. If maximum value of v so that it does not strike to sphere is $\frac{1}{8}\sqrt{\frac{46Qq}{xmR\pi\varepsilon_0}}$, then find x. (Neglect gravity). (Both charges are of same nature)



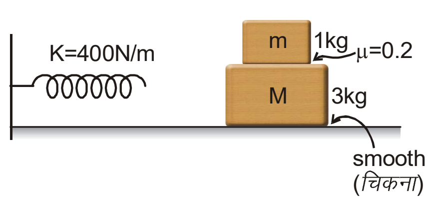
15. Figure shows cone of mass 200 gm, suspended by a vertical jet of water from a garden hose. The flow rate is 0.2 kg/sec. The water rises 8m if unimpeded. Find the height of the cone top surface (h in m) at which water strike. Assume the water emerges horizontally from holes at the top of cone.



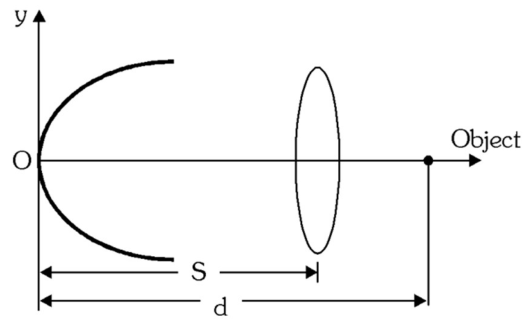
16. A hollow cone made of insulating material has a total charge Q = 1 μC spread uniformly over its sloping surface. Find the work (in Joule) to take a test charge q = 1 mC from infinity to apex of the cone. Slant length of the cone is ℓ = 6 m.

17. Light from a sodium lamp λ = 590 nm is diffracted by a slit of width d = 0.30 mm. The distance from the slit to the screen is D = 0.87 m. Find the width of the central maximum.

18. As shown in the figure, there is no friction between the horizontal surface and the lower block (M = 3 kg) but friction coefficient between both the blocks is 0.2. Both the blocks move together with initial speed V towards the spring, compresses it and due to the force exerted by the spring, moves in the reverse direction of the initial motion. The maximum value of initial speed V (in cm/s) so that during the motion, there is no slipping between the blocks is x, then value of $\frac{x}{4}$ is: (use g = 10 m/s^2).



19. A point object is kept at a distance d = lm from a parabolic reflecting surface $x = 8y^2$. An equiconvex lens is kept at a distance s = 80 cm from the parabolic surface (see figure.). The focal length of the lens is 20 cm. Find the position of the image after reflection from the surface.



20. Characteristic X–rays of frequency 4.2×10^{18} Hz are produced when transitions from L-shell, K-shelltake place in a certain target material. Use Mosley's Law to determine the atomic number of the target material. Given Rydberg constant $R = 1.1 \times 10^7\ m^{-1}$.

ANSWER KEY

1.	(C)	**2.**	(A)	**3.**	(A)	**4.**	(C)	**5.**	(A)
6.	(D)	**7.**	(BD)	**8.**	(A,B)	**9.**	(ABC)	**10.**	(A,B)
11.	(C)	**12.**	(A)	**13.**	(B)	**14.**	2	**15.**	3
16.	3	**17.**	3.4	**18.**	5	**19.**	3.16	**20.**	42